



CHEMISTRY Department
SIMHADRI

Note No :1



Ref:SMTTP/Chemistry/2024-25/LMI rev/661269

Date:

11/11/2024(11:21:38)

Subject:CHEMICAL CONTROL OF GENERATOR STATOR PRIMARY
WATER SYSTEM

TheLMI/OGN/OPS/CHEM/002 on Chemical Control of Generator Stator Primary Water System pertaining to O&M (CHEM) section has been reviewed and revised. Changes made are mentioned in the below table.

S. no	clause	Revision 8	Revision 9
1.	6.0	-	New line added: The sample line should have two number isolating valves.
2.	7.1	-	An online DO analyzer shall be installed in Low-DO Generator Stator Water System for continuous monitoring of the dissolved oxygen in the stator primary water
3.	9.2.3	-	Whenever N2 gas pressure drops below <0.10 Kg/cm2 then N2 gas top-up/pressurization is to be done
4.	9.2.12	-	Spare resin strainers should be available for SPU system

5.	9.2.16	-	Feedback of NaOH dosing pump should be available in unit control room
6.	9.2.18	Ion exchanger (SPU) outlet conductivity trend & Stator water Conductivity are also to be monitored and recorded in Generator stator water logbook once/shift	Ion exchanger (SPU) outlet conductivity trend & Stator water Conductivity are also to be monitored and recorded in Generator stator water logbook once/day
7.	9.2.24.2	Soda lime filter of NaOH tank should be changed as per OEM recommendation.	Alkali tank soda filter is to be replaced, Once/year
8.	10.4	-	Added: The stator water sample to be cross-checked through IC/ICP/AAS, Once/six months for Total Copper
9.	10.5	-	Added: Soda lime filter changeover & record Responsibility: Electrical
10.	Annexure 1. Note:	CPU resins (Cation in -H form and Anion in -OH form) should be used in the SPU system.	CPU resins (Cation in -H form and Anion in -OH form) should be used in the SPU system. However, if the station is facing any issue in the SPU performance with CPU resins, then station should use only OEM recommended resin (Refer: Generator O&M manual for details)
11.	Annexure 1	-	Added: Generator stator water flow, winding DP, SPU flow and N2/ H2 gas pressure are also to be recorded in the 'Generator stator water chemistry logbook' along with the above results.

- | | | | |
|-----|------------|---|--|
| 12. | Annexure 2 | - | Added Signature of Chemistry and Operation in Annexure-2 |
| 13. | Annexure-5 | - | Added standard PI tag list for Stator primary water system in Annexure-5 |

Submitted for kind approval please.

Submitted by:

मयंक बिष्ट - सहायक प्रबंधक

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On:11/11/2024(11:21:38) **Ref:**SMTTPP/Chemistry/2024-25/LMI rev/661269

Note No:2

Based on the revision of original OGN, the LMI on the above topic is reviewed and revised. All relevant changes are made in the LMI and listed accordingly.

All the concerned departments are requested to review the document and comment if any.

submitted for reviewing the document and subsequent approval by Competent Authority please.

Submitted by:

सांतनु जेना - उप महाप्रबंधक (रसायन)

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On:11/11/2024(18:41:57) **Ref:**SMTTPP/Chemistry/2024-25/LMI rev/661269

Note No:3

1. In Clause 9.2 'Monitoring & Control', following changes may be done:

Sl. No.8: add HOD (MM-OS) in place of HOD (AHD/CPC),

Sl. No.11: add HOD (C&I)

Sl. No.16: add HOD (C&I)

Sl. No.24.2: add HOD (TMD)

2. In LMI approval document add HODs of EMD, MM-OS, C&I & TMD in cheking.

Submitted by:

सोमनाथ भट्टाचार्य - अपर महाप्रबंधक

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On:13/11/2024(19:43:12) **Ref:**SMTPP/Chemistry/2024-25/LMI rev/661269

Note No:4

LMI revised accordingly. Submitted for approval

Submitted by:

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On:18/11/2024(17:25:01) **Ref:**SMTPP/Chemistry/2024-25/LMI rev/661269

Note No:5

Submitted for approval please.

Submitted by:

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On:20/11/2024(11:23:23) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:6

Checked and forwarded.

Submitted by:

सोमनाथ भट्टाचार्य - अपर महाप्रबंधक

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On:22/11/2024(16:15:54) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:7

Please review.

Submitted by:

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On:23/11/2024(15:09:59) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:8

reviewed

Submitted by:

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On:23/11/2024(15:44:29) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:9

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On:26/11/2024(10:08:20) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:10

May review 10.5 Soda lime filter changeover and record keeping responsibility.

Additional points to be added.

Fixing of responsibility of work and recordkeeping NaOH dosing tank level makeup and concentration of solution.

Submitted by:

बाला बालाजी आकुला - उप महाप्रबंधक

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On:02/12/2024(10:33:41) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:11

10.5 corrections done

record keeping of NaOH tank level is mentioned in 9.2.17

recommended Caustic concentration is mentioned in annexure 4

Submitted by:

मयंक बिष्ट - सहायक प्रबंधक

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On:17/12/2024(11:21:10) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:12

NaOH dosing tank soda lime filter chemical compound shall be supplied by chemistry and replacement shall be done by EM as being followed in replacement of SPU resin.

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On:21/12/2024(12:56:58) **Ref:**SMTTPP/Chemistry/2024-25/LMI rev/661269

Note No:13

Submitted by:

तेन्नेटि विजय कृष्ण - अपर महाप्रबंधक

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On:21/12/2024(16:14:19) **Ref:**SMTTPP/Chemistry/2024-25/LMI rev/661269

Note No:14

The changes are made in the LMI accordingly. May please process the proposal.

Submitted by:

सांतनु जेना - उप महाप्रबंधक (रसायन)

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On:23/12/2024(11:34:09) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:15

the LMI is reviewed and found in order wrt EM.

Submitted by:

बाला बालाजी आकुला - उप महाप्रबंधक

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On:23/12/2024(12:06:38) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:16

Submitted by:

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On:23/12/2024(19:41:53) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:17

Reviewed and is being forwarded for further process please.

Submitted by:

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On:26/12/2024(15:22:14) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:18

May pl review the LMI document

Submitted by:

जे आर वर्गीस - अपर महाप्रबंधक

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On:26/12/2024(18:41:46) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:19

Submitted by:

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On:26/12/2024(19:23:20) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:20

LMI for Chemical Control of Generator Stator Primary Water System [chemical control of primary water system FINAL.pdf](#) reviewed.

"An online DO analyzer shall be installed in Low-DO Generator Stator Water System for continuous monitoring of the dissolved oxygen in the stator primary water"

Forwarded for kind approval.

Submitted by:

प्रदीप कुमार मौर्य - वरिष्ठ प्रबंधक

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On:07/01/2025(10:49:25) **Ref:**SMTTPP/Chemistry/2024-25/LMI rev/661269

Note No:21

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On:09/01/2025(20:28:04) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:22

Forwarded after review

Submitted by:

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On:11/01/2025(19:38:22) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:23

Please review

Submitted by:

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On:15/01/2025(20:45:47) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:24

As discussed may please review responsibility against clause no. 24.2

Submitted by:

के पी रमेश - वरिष्ठ प्रबंधक

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On:16/01/2025(19:13:24) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:25

LMI Revised as discussed.

Submitted by:

मयंक बिष्ट - सहायक प्रबंधक

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On:27/01/2025(18:43:52) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:26

Submitted by:

सांतनु जेना - उप महाप्रबंधक (रसायन)

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On:27/01/2025(18:51:46) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:27

Attached LMI "chemical control of primary water system FINAL.pdf" reviewed.

Forwarded for approval.

Submitted by:

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On:27/01/2025(19:11:27) **Ref:**SMTTPP/Chemistry/2024-25/LMI rev/661269

Note No:28

forwarded for approval please

Submitted by:

नीलांजन मंडल - अपर महाप्रबंधक

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On:28/01/2025(10:04:03) **Ref:**SMTTPP/Chemistry/2024-25/LMI rev/661269

Note No:29

Pls review sl no 8 , page no 11.

Also any additions if required, to be done.

Submitted by:

टी वी राजशेखर - अपर महाप्रबंधक

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On:10/02/2025(10:23:09) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:30

LMI Reviewed & Clause no. 9.2.8 is being complied as per LMI.

Submitted by:

हेमन्त सैनी - उप महाप्रबंधक

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On:10/02/2025(11:07:34) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:31

Forwarded please.

Submitted by:

टी वी राजशेखर - अपर महाप्रबंधक

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On:11/02/2025(19:02:49) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:32

As discussed, change in page number-12 to be done.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

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On:12/02/2025(10:42:59) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:33

As discussed, the document is edited and copy attached.

Submitted for approval please.

Submitted by:

सांतनु जेना - उप महाप्रबंधक (रसायन)

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On:12/02/2025(16:12:17) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:34

LMI is in line with the OGN, may please be approved

Submitted by:

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On:13/02/2025(09:39:20) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:35

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On:13/02/2025(16:33:46) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

Note No:36

Approved & Submitted by:

समीर शर्मा - मुख्य महाप्रबंधक

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On:13/02/2025(23:12:33) **Ref:**SMTTP/Chemistry/2024-25/LMI rev/661269

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

CHEMICAL CONTROL OF GENERATOR STATOR PRIMARY WATER SYSTEM

LMI/OGN/OPS/CHEM/002, Revision: 09

N o v -2024

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	Chemical Control of Generator Stator Primary Water System
2	LMI No.	LMI/OGN/OPS/CHEM/002 , Revision: 09
3	Reference Document	COS-ISO-00-OGN/OPS/CHEM/002 Rev. No.: 06, Oct. 2024
4	Superseded Document	LMI/OGN/OPS/CHEM/002 Issue: 01, Revision: 08 SEP 2022
5	Prepared By	Name: Mayank Bisht, Asst Manager (Chem)
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7	Reviewed By	Name: Prasanta Kumar Jena GM (O&M)
8	Approved By	Name: Sameer Sharma H O P (Simhadri)

TITLE	Chemical Control of Generator Stator Primary Water System
LMI No.	LMI/OGN/OPS/CHEM/002- Rev. No-09 - Nov-2024

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TITLE	Chemical Control of Generator Stator Primary Water System
LMI No.	LMI/OGN/OPS/CHEM/002- Rev. No-09 - Nov-2024

1.0 INTRODUCTION

In early generator technology, heat generated inside the generator was removed by gas cooling, either with air or with nitrogen. Water cooling was introduced in 1956 in order to accommodate larger generator size.

The directly cooled generators are provided with roebel bar hollow conductors connected to common water headers at inlet and outlet of the winding. The conductors are made of copper and cooling water header and piping are made of stainless steel. The generator stator cooling water system consists of heat exchangers, circulating pumps for water, fans for gas, filters, and connecting features. Cooling water is purified by ion exchange process and its dissolved oxygen is controlled either to very low levels (< 20ppb) or to high levels (> 2ppm) as per manufacturer's guidelines.

The monitoring and control of primary water quality is essential to maintain the electrical insulation of the machine, to prevent corrosion and deposition of corrosion products in stator water circuit and to eliminate the risk of obstructing the cooling water flow through the stator bars, and to minimize the maintenance.

In Simhadri as per the manufacturer's guidelines only low oxygen regime is being followed.

2.0 SCOPE:

This LMI deals with Chemical control limits of Stator Primary Water in the units of Simhadri Thermal Power Station. This LMI also elaborates on guide lines for make up to primary water, specifications for makeup, and purge gas Nitrogen.

3.0 SUPERSEDED DOCUMENTS:

LMI/OGN/OPS/CHEM/002

Issue: 01, Revision: 08

SEP 2022

TITLE	Chemical Control of Generator Stator Primary Water System
LMI No.	LMI/OGN/OPS/CHEM/002- Rev. No-09 - Nov-2024

4.0 RECORDS OF REVISION

Recommended chemical control limits for generator stator primary water is based on the revised OGN, COS-ISO-00-OGN/OPS/CHEM/002 Rev. No.: 06, Oct. 2024.

Current revision is carried out based on latest modifications made in the OGN

S. No	clause	Revision 8	Revision 9
1.	6.0	-	New line added: The sample line should have two number isolating valves.
2.	7.1	-	An online DO analyzer shall be installed in Low-DO Generator Stator Water System for continuous monitoring of the dissolved oxygen in the stator primary water
3.	9.2.3	-	Whenever N2 gas pressure drops below <0.10 Kg/cm2 then N2 gas top-up/pressurization is to be done
4.	9.2.12	-	Spare resin strainers should be available for SPU system
5.	9.2.16	-	Feedback of NaOH dosing pump should be available in unit control room
6.	9.2.18	Ion exchanger (SPU) outlet conductivity trend & Stator water Conductivity are also to be monitored and recorded in Generator stator water logbook once/shift	Ion exchanger (SPU) outlet conductivity trend & Stator water Conductivity are also to be monitored and recorded in Generator stator water logbook once/day
7.	9.2.24.2	Soda lime filter of NaOH tank should be changed as per OEM recommendation.	Alkali tank soda filter is to be replaced, Once/year
8.	10.4	-	Added: The stator water sample to be cross-checked through IC/ICP/AAS, Once/six months for Total Copper
9.	10.5	-	Added: Soda lime filter changeover & record Responsibility: Electrical

TITLE	Chemical Control of Generator Stator Primary Water System
LMI No.	LMI/OGN/OPS/CHEM/002- Rev. No-09 - Nov-2024

10.	Annexure 1. Note:	CPU resins (Cation in -H form and Anion in -OH form) should be used in the SPU system.	CPU resins (Cation in -H form and Anion in -OH form) should be used in the SPU system. However, if the station is facing any issue in the SPU performance with CPU resins, then station should use only OEM recommended resin (Refer: Generator O&M manual for details)
11.	Annexure 1	-	Added: Generator stator water flow, winding DP, SPU flow and N2/ H2 gas pressure are also to be recorded in the 'Generator stator water chemistry logbook' along with the above results.
12.	Annexure 2	-	Added Signature of Chemistry and Operation in Annexure-2
13.	Annexure-5	-	Added standard PI tag list for Stator primary water system in Annexure-5

5.0 REQUIREMENTS FOR DIRECT COOLING OF WINDING CONDUCTORS WITH WATER

- The conductivity of the water must be very low to prevent current flow and electrical flashover.
- The velocity of water must be low enough to prevent erosion. The water should be treated to minimize Corrosion. Either of these two can lead to deposition of erosion/corrosion products within the conductor waterways and the risk of obstructing the cooling water flow through the bars.
- The maximum water pressure must be lower than the casing hydrogen pressure so that if any leakage occurs it is of hydrogen into the water circuit since leakage of water into the winding insulation could lead to an electrical breakdown.
- The maximum temperature in the water circuit must be low enough to provide an adequate margin below boiling point. The inlet temperatures are generally just above 40°C and outlet temperatures 65-70°C
- The means used to transfer water into the conductors must be of high integrity

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insulation material e.g., PTFE, poly tetra fluoroethylene, not easily degraded.

6.0 GENERAL DESCRIPTION OF THE SYSTEM OF STATOR WATER SYSTEM

The primary water required for cooling the live components is circulated in a closed system. In order to prevent corrosion materials like copper, stainless steels or similar corrosion resistant materials are used in the entire cooling water system. The primary water system comprises of the following principal components:

- Primary water tank
- Primary water pumps
- Filters
- Primary water treatment system
- Alkalizer unit

The primary water admitted to the pump from the primary water tank is passed to the Exciter End cooling water manifold in the generator interior and then to neutral point of the terminal bushings via the cooler and the filters. After having absorbed the heat losses in the stator windings, phase connectors and main bushings, the water is returned to the primary tank from the Turbine End cooling water manifold or the phase connectors. The gas pressure above the water level in the primary water tank is maintained constantly by a pressure controller.

Primary Water Tank

The primary water tank is arranged in an easily accessible location on the stator frame on an elastic support, thus forming the highest point on the entire primary water system in terms of static head. Nitrogen capping is provided in the space above water level in primary water tank at a constant pressure by a pressure controller to avoid atmospheric air coming in contact with primary water. Nitrogen capping also facilitates deaeration.

Primary Water Pump

The primary water is permanently circulated in a closed system. To ensure uninterrupted generator operation, two full-capacity pumps are provided. Each pump is driven by a separate motor. In the event of failure of one pump, the stand by pump starts automatically.

Filters

Filters are provided in the suction as well as discharge of primary water pump to prevent blockage in stator water circuit by foreign material.

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Primary water treatment system

In order to maintain such a low conductivity, a small quantity of stator water (around 2% of total flow) is continuously passed through an ion exchanger which is an integral part of the primary water system. The ion exchange resins used in stator water polishing unit can be replaced while the generator is on load.

Alkalizer unit

The low oxygen primary water is alkalized to prevent flow blockage due to copper corrosion in water cooling windings. The pH of the primary water is maintained around 8.5 to 9.0 by continuous dosing of dil. NaOH solution (0.5 – 1.0 % w/v) to prevent corrosion of copper conductors in the circuit . Corrosion products may lead to blockage in the windings.

Water Analysis

It is recommended that chemical analysis of primary water in the main system and water treatment system is carried out (downstream of fine filter) at monthly intervals for the following parameters: Conductivity, pH, Dissolved Oxygen, Cu and Fe. The stator primary water sample should be taken from the generator winding inlet header only. The sample line should have two number isolating valves.

7.0 CORROSION OF COPPER IN STATOR COOLING WATER SYSTEM

There are two regimes where the copper is best passivated and minimum corrosion is found. These are:

1. Low oxygen regime (< 20ppb) where corrosion is inhibited by formation of passive layer of cuprous oxide i.e. Cu_2O . This regime is adopted one for machines of 500 mw and above. Primary water system for high capacity generators are designed with Nitrogen capping provision. The space above water in the primary water tank is filled with high purity nitrogen. Nitrogen capping facilitates de- aeration and acts as a sealing from atmospheric air. These features are in line with low oxygen operating regime. An online DO analyzer shall be installed in Low-DO Generator Stator Water System for continuous monitoring of the dissolved oxygen in the stator primary water.
2. High oxygen regime (> 2ppm) where corrosion is inhibited by formation of passive layer of cupric oxide i.e. CuO . This regime is adopted for 200 MW capacity and less .Primary water system for low capacity generators are designed to operate with high oxygen regime where there is no requirement of de-aeration or nitrogen capping.

Variation of oxygen level from one regime to other leads to accelerated corrosion and

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dislodge copper oxide scale which could cause blockage of primary water circuit in the stator water conductor and may lead to serious failures. Hence dissolved oxygen level is to be maintained within limits as per the selected operating regime. For all practical purpose, shifting of operating regimes is not recommended.

Further the amount of copper deposits in the primary water system profoundly depends on :

- a) Quality and quantity of makeup water
- b) Concentration of dissolved oxygen
- c) Quality and pressure of nitrogen gas in stator primary water tank
- d) Operation of alkalizer system
- e) Operation of stator water polishing unit
- f) Primary stator water cooler leakage (ingress of impurity)

8.0 SOURCE OF MAKEUP WATER

Source of makeup water for the stator water cooling is DM water from the water treatment plant. This water has a specific conductivity of less than 1.0 $\mu\text{S}/\text{cm}$, pH about 6.0-7.5, and a high dissolved oxygen content of about 6 to 8 ppm depending on water temperature.

9.0 SPECIFIC INSTRUCTIONS TO BE FOLLOWED

Source of makeup water for the stator water cooling is DM water from the water treatment plant. This water has a specific conductivity of less than 1.0 $\mu\text{S}/\text{cm}$, pH about 6.0-7.5, and a high dissolved oxygen content of about 4 to 8 ppm depending on water temperature.

S.no	Activity	Responsibility
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9.1	PROCEDURE FOR TAKING MAKEUP INTO THE STATOR WATER SYSTEM	
1.	Make up to the stator water system should preferably be taken during unit shut down after bypassing the stator windings and degassing the water using nitrogen purging. In emergency, make up should be taken in the system in small quantity (< 10 liters at a time). Record should be maintained for makeup water taken in the system. During make up purging with nitrogen in primary water tank should be ensured to bring down the dissolved oxygen content to possible lowest value (about 20 ppb)	SCE /Unit Control Engr/ HOD (EMD)
2.	The DM water should be checked for the parameters given in Annexure 3 before taking into stator water system.	HOD(Chem)
3.	When unit is in operation, makeup should be taken in the system in low quantities and in stages and in no case more than 10 liters of makeup is to be taken at a time.	SCE /Unit Control Engr
4.	A joint protocol with electrical, operation, and chemistry should be made for taking any makeup into the stator water system indicating the quantity, quality, and stator water parameters.	SCE /Unit Control Engr/ HOD (EMD)/ HOD (Chem)
9.2	MONITORING AND CONTROL	
1.	Monitoring of alkalizer pump operation, SPU flow, SPU outlet conductivity, Winding inlet and outlet Conductivity and all parameters should be recorded in Shift logbook once/shift.	SCE /Unit Control Engr
2.	It is necessary to regulate the makeup water flow in a manner that dissolved oxygen does not abruptly increase. It is advisable that makeup should only be taken into system with winding in bypass condition.	SCE /Unit Control Engr
3.	Dissolved oxygen in stator cooling water is generally removed by purging with Pure grade nitrogen gas (99.999%, BIS: 1747-1972). Removal of dissolved oxygen depends on the nitrogen pressure which is normally maintained at around 0.3 kg/cm ² with low alarm for nitrogen pressure in primary water tank set at 0.1 kg/cm ² and high alarm set at 0.5 kg/cm ² for better efficiency. Whenever N ₂ gas pressure drops below <0.10 Kg/cm ² then N ₂ gas topup/pressurization is to be done.	SCE /Unit Control Engr

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4.	A bypass ion exchange polishing plant is installed to control the conductivity of the stator water and can also trap some of the suspended matter in water. Polishing plant is generally designed to treat about 2-5% of total stator water flow. The polisher containing a bed of mixed resins consisting of strong Cation resin in H-form and strong anion resin in OH form should be kept in service.	Unit Control Engr SCE/ HOD(Chem)
5.	Under the low oxygen regime corrosion of copper is retarded by minimizing the oxygen concentration. Once in a month stator primary water shall be tested for conductivity, pH, and Dissolved oxygen and copper	HOD (Chemistry)
6.	In an iron free system, the iron content should be mainly due to iron entering via the makeup water. The test should be done for iron monthly to confirm the absence of mild steel in a nominally steel free system.	HOD (Chemistry)
7.	More frequent analysis of copper, iron, and pH may be required when the unit is being commissioned or when the recommended limits are violated.	HOD (Chemistry)
8.	It must be ensured that the DM water storage tanks providing makeup to stator primary water system should be coated with epoxy coating material. The inspection of lined surfaces should be carried out periodically and corrective measures be taken immediately on detection of any failure of lining.	HOD (MM- OS) AGM(Chemis try)
9.	The rate of corrosion is minimum when the stator primary water system is tight i.e. when neither water is lost from the system nor the makeup is taken into the system. Practically, this is not possible as water is taken out for manual testing of the constituents and sometimes lost due to leakages. It is desirable that manual tests are carried out once in a month, so that minimum water is taken out from the system.	HOD (Chemistry)
10.	The makeup is taken on certification of quality by Chemistry it is jointly recorded. Reason for any abnormal increase in normal make up is to be investigated and reported. N2 purging is also simultaneously done when makeup is being taken in the outer circuit of the stator windings.	SCE /Unit Control Engr/ HOD (EMD)/ HOD (Chem)

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11.	The corrosion products in stator primary water when detached from the metal surfaces, deposit on the tube ends and also on inlet and outlet hoses of water cooled generator windings, resulting in flow rate restriction. In order to get timely indication of flow restriction, it is essential to provide individual pressure gauges at inlet and outlet of water flowing through stator windings. Differential pressure indicator gauge and transducer (for UCB indication) with alarm should be monitored for avoiding long shut down of unit.	SCE /Unit Control Engr/ HOD (C&I)
12.	During cold start up condition, strainer deposits should be analyzed for oxides of copper for assessing the loose oxide coat getting dislodged. Spare resin strainers should be available for SPU system	HOD (Chemistry)/ HOD (EMD)
13.	Ingress of leaking hydrogen into water will reduce dissolved oxygen level Generator should not run on Hydrogen gas leakages beyond the period as recommended by the manufacturers.	SCE /Unit Control Engr
14.	DP and Generator temperature rise should be closely monitored. Trending of DP per unit flow to be carried out for detecting any change in DP on monthly basis. Differential pressure, winding flow and generator slots temperature rise should be closely monitored.	SCE /Unit Control Engr
15.	Dissolved oxygen and conductivity of stator primary water system should not vary frequently.	HOD(Operati on)&HOD (Chem)
16.	Monitoring of alkalizer pump operation, SPU flow, SPU outlet & Stator water Conductivity etc. Feedback of NaOH dosing pump should be available in unit control room.	SCE /Unit Control Engr HOD (C&I)
17.	Monitoring of NaOH tank level, minimum once in a day.	SCE /Unit Control Engr
18.	Ion exchanger (SPU) outlet conductivity trend & Stator water Conductivity are also to be monitored and recorded in Generator stator water logbook once/day	Shift chemist through PI
19.	Offline checking of all chemistry parameters as per schedule specified in different annexures.	HOD (Chem)
20.	Chemical Control Limits of Stator Primary water as given in annexure-I should be ensured	HOD(Operati on)&HOD (Chem)

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21.	All registers/ log sheets should be perused every month end and preserved	HOD(Operati on)&HOD (Chem)
22.	Special Stator water register/log sheet kept with the shift charge engineer shall contain details about leaks and other losses observed in stator water system, make up water quality and quantity, details of SPU resin change and details of Nitrogen purging. It shall contain Stator water temperature, Inlet Pressure, differential pressure, Specific conductivity, Nitrogen capping pressure, and pump in service on a shift wise basis.	SCE /Unit Control Engr
23.	All exceptions and deviations should be reported to HOD (O) and HOD(Chemistry) who will dispense the deviations.	SCE /Shift Chemist
24.	ALKALIZER SYSTEM CONTROL	
24.1.	The limiting values low oxygen regimes are pH dependent and it is important that the pH is controlled and monitored. An alkalizer is used for maintaining the desired pH values for minimizing corrosion. Alkalizer should be operated to maintain a pH of 8.5 to 9.0 and specific conductivity of 1.8 to 2.2 in Stator Primary water.	SCE /Unit Control Engr /HOD(C&I)
	AR grade sodium hydroxide should be used for preparation of caustic solution. DM water (conductivity value <1.0 $\mu\text{S}/\text{cm}$) should be used for preparation of caustic solution.	HOD (Chem)
24.2.	A protocol (Ref: Annexure-4) for NaOH solution preparation is to be prepared and signed by the Chemistry & Operation which will ensure that solution is prepared properly. During unit overhauling, cleaning of NaOH tank should be done. Alkali tank soda filter is to be replaced, Once/year.	HOD(Operati on)/HOD (Chem)/
24.3.	Same concentration of sodium hydroxide solution should always be used for top up in NaOH tank of alkalizer unit to avoid any variation in stator primary water conductivity.	HOD (Chem)

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10.0 RECORDS TO BE MAINTAINED

Sl. No.	Record description	Maintained by	Frequency
1.	Offline analysis records of Stator water and makeup Water	Chemistry	As specified in annexures
2.	All specified monitoring parameters	Chemistry/ Operation/ Electrical	As specified in diff sections
3.	SPU changeover and caustic top up records	Chemistry	As specified in Annexure-4,5
4.	Stator water copper checking through IC/ICP/AAS-record	Chemistry	Once/6-months
5.	Soda lime filter changeover & record (Chemical will be supplied by Chemistry and replacement will be done by EMD)	Electrical/Chemistry	Once/ year

11.0 REFERENCE DOCUMENT:

COS-ISO-00-OGN/OPS/CHEM/002 Rev. No.: 06, Oct. 2024

12.0 REVIEW

Head of O&M is responsible for reviewing the document on a 2-yearly basis and as required.

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ANNEXURE-1

**RECOMMENDED CHEMICAL CONTROL LIMITS FOR GENERATOR STATOR PRIMARY
WATER OF 500 MW UNITS**

Sl. No.	PARAMETER	CONTROL LIMITS (WITH ALKALIZER)	FREQUENCY OF TESTING
1	Conductivity at 25°C, μS/cm	1.8-2.2	Online/Continuous
2	pH at 25°C	8.5-9.0	Once in month
3	Dissolved oxygen, PPB	<20	Once in month Online continuous if analyzer is provided
4	Total copper, PPB	<10	Once in month
5	Total iron, PPB	< 5	Once in month
6	Total sodium, PPB	25-250 (max. 300)	Once in month

i Testing frequency may be increased if any of the above parameter is maintaining on higher side.

ii Generator stator water flow, winding DP, SPU flow and N₂/ H₂ gas pressure are also to be recorded in the 'Generator stator water chemistry logbook' along with the above results. Frequency as mentioned above.

iii The stator water sample is to be cross-checked through IC/ICP/AAS Once/6- months for total copper

iv For Coastal units, chloride also to be checked along with above parameters in stator primary water.

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**CONDUCTIVITY ($\mu\text{S}/\text{cm}$) DETAILS FOR UNITS PROVIDED WITH ALKALIZER UNIT
FOR MAIN CIRCUIT AFTER ION EXCHANGER**

AFTER ION EXCHANGER CONDUCTIVITY at 25°C (for Alkalizer Pump Start/ Stop)	
Pump Start , $\mu\text{S}/\text{cm}$	$\leq 1.0 \mu\text{S}/\text{cm}$
Pump Stop , $\mu\text{S}/\text{cm}$	$>3.0 \mu\text{S}/\text{cm}$ (or Conductivity after main filter $>2.5 \mu\text{S}/\text{cm}$)

Note:

- i. Any change in alkalizer pump start & stop logic/ set points should be avoided.
- ii. When SPU outlet conductivity is not reaching $\leq 1.0 \mu\text{S}/\text{cm}$ in NaOH pump stop condition, then it should be declared exhausted.
- iii. The SPU resins must be replaced with ***new regenerated cation and anion resins*** to get maximum output. It will also minimize SPU changeover frequency and subsequently exposure of generator winding with high DO water.
- iv. As there is possibility of cross-contamination during separation of exhausted mixed resins, so these resins should not be reused in SPU again.
- v. CPU resins (Cation in -H form and Anion in -OH form) should be used in the SPU system. However, if the station is facing any issue in the SPU performance with CPU resins, then station should use only OEM recommended resin (Refer: Generator O&M manual for details).
- vi. Before charging of fresh resins in SPU, both cation and anion should be separately regenerated and rinsed to get conductivity $<1.0 \mu\text{S}/\text{cm}$.
- vii. After changeover of exhaust resins, the SPU should be thoroughly rinsed with DM water to get rinse conductivity $<0.30 \mu\text{S}/\text{cm}$ (online/offline) at outlet of ion exchanger & fine filter. The offline measurement should be performed with the sample excluded from the atmosphere to prevent a falsification of the reading by the carbon dioxide contained in the ambient air.
- viii. Portable conductivity meter should be used for offline testing of water samples at local.
- ix. Ensure that alkalizer pump must be in isolated condition during changeover of ion exchanger resin and when SPU is not in service.
- x. A joint protocol is to be maintained as per Annexure-3.

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ANNEXURE-2**MAKE UP WATER (DM WATER) SPECIFICATIONS**

SP. CONDUCTIVITY, $\mu\text{S}/\text{cm}$	<1.0
pH	6.0 – 7.5
DISSOLVED OXYGEN, ppm	6 - 8
TOTAL COPPER, ppb	: <5
TOTAL IRON, ppb	:<5

Name & Signature:**O&M (Chemistry)****O&M (Operation)**

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ANNEXURE-3

**PROTOCOL FOR RECORD OF TAKING MAKE UP WATER IN STATOR PRIMARY WATER
SYSTEM/SPU RESIN REPLACEMENT**

Unit:**Date:****Unit Status:** Onbar/offbar

SL. NO.	CONTENT	INITIAL	FINAL (After makeup taken / SPU changeover)	REMARKS
1.	Stator primary water tank level (mm)			
2.	Makeup water totalizer reading (Ltr/m ³)			
3.	Stator primary water conductivity – Winding inlet (μS/cm)			
4.	Stator primary water conductivity – Winding outlet (μS/cm)			
5.	Ion exchanger rinse conductivity- Online & Lab(μS/cm),when SPU change over done	----		
6.	Dissolve Oxygen in stator water system (ppb)			
7.	Quantity of makeup water taken for system makeup (ltr.):			

Note:

Name & Signature:**(Chemistry)****(Operation)****(Elect. Maintenance)**

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ANNEXURE-4

PROTOCOL FOR CAUSTIC TOPUP IN NaOH TANK OF ALKALIZER

Date:

Unit:

Status: On bar/off bar

SL. NO.	CONTENT	Required	Initial value	Final value	REMARKS
1.	DM water conductivity	< 1uS/cm	--	--	
2.	Concentration of caustic solution topped up	0.5-1.0% (w/v)	--		
3.	Quantity of caustic solution used in top up	--	--		
4.	Level of NaOH tank	--			

Name & Signature:

(Chemistry)

(Operation)

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Annexure-5:**STANDARD PI TAG LIST FOR MONITORING OF STATOR WATER PARAMETERS**

Sl. No.	Parameter
1	Generator winding inlet conductivity
2	Generator winding outlet conductivity
3.	Ion exchanger (SPU) outlet conductivity
4.	Ion exchanger flow (if flow transmitter available)
5.	Stator primary water DO
6.	Stator primary water flow main winding
7.	Stator primary water DP across winding
8.	Primary water tank-N2 gas pressure for Low DO regime generator
9.	Stator primary water tank level

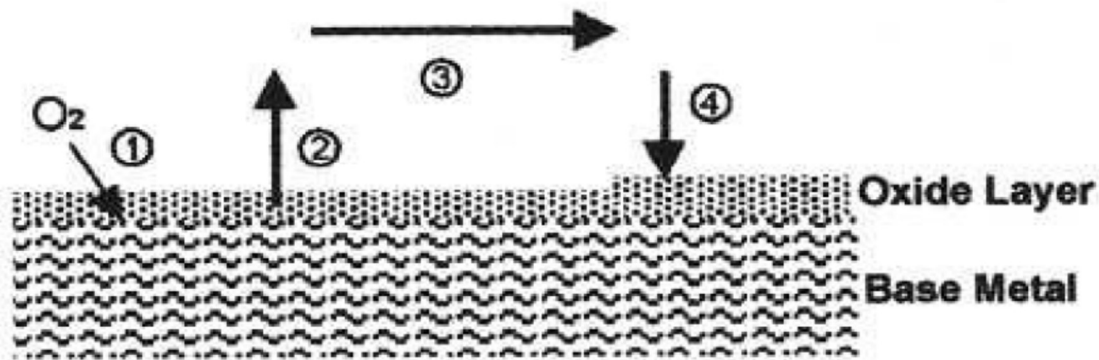
ANNEXURE-6**EXTRA PURE NITROGEN GAS SPECIFICATIONS FOR STATOR PRIMARY WATER SYSTEM
(BIS 1747-1972)**

The maximum impurity level in N2 gas used for stator primary water system shall be as follows:

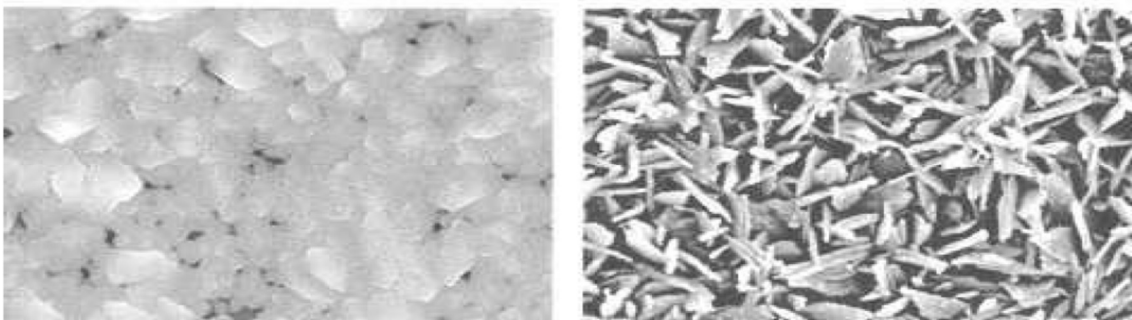
S. No.	PARAMETER	UNIT	Limits FOR EXTRA PURE GRADE 1
1.	Oxygen, Max	ppm	2.0
2.	Moisture, Max	ppm	2.0
3.	Carbon dioxide, Max	ppm	Nil
4	Carbon monoxide, Max	ppm	Nil
5	Oxides of nitrogen, Max	ppm	Nil
6	Hydrogen, Max	ppm	Nil
7	Argon, Max	ppm	5.0
8	Hydrocarbons, Max	ppm	0.2
9	Sulphur contents, Max	ppm	Nil
10	Mercury, Max	ppm	Nil
11	Chlorine, Max	ppm	Nil

The above parameters are to be confirmed from the test certificate of gas by OEM/NABL lab.

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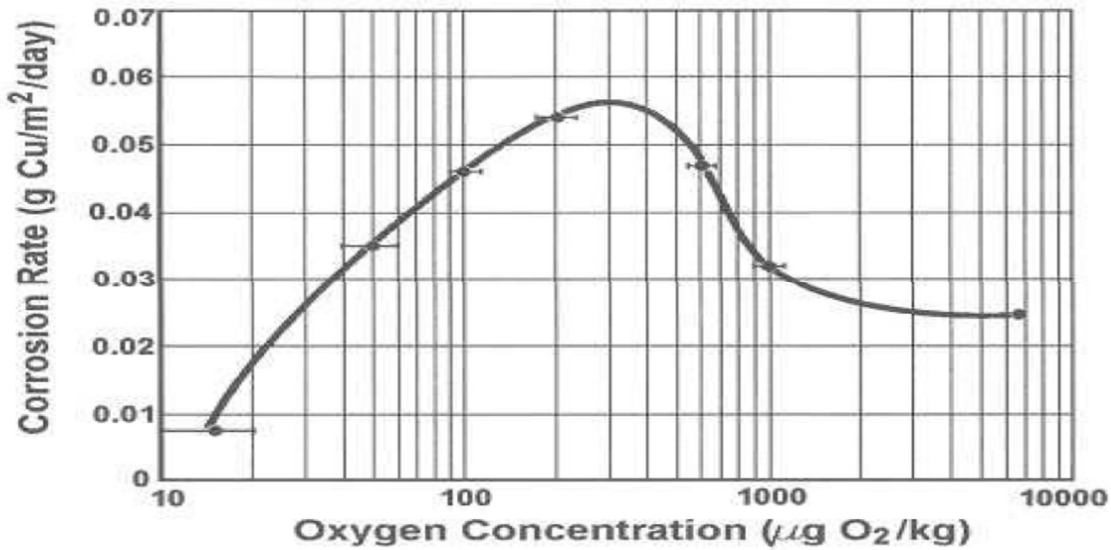
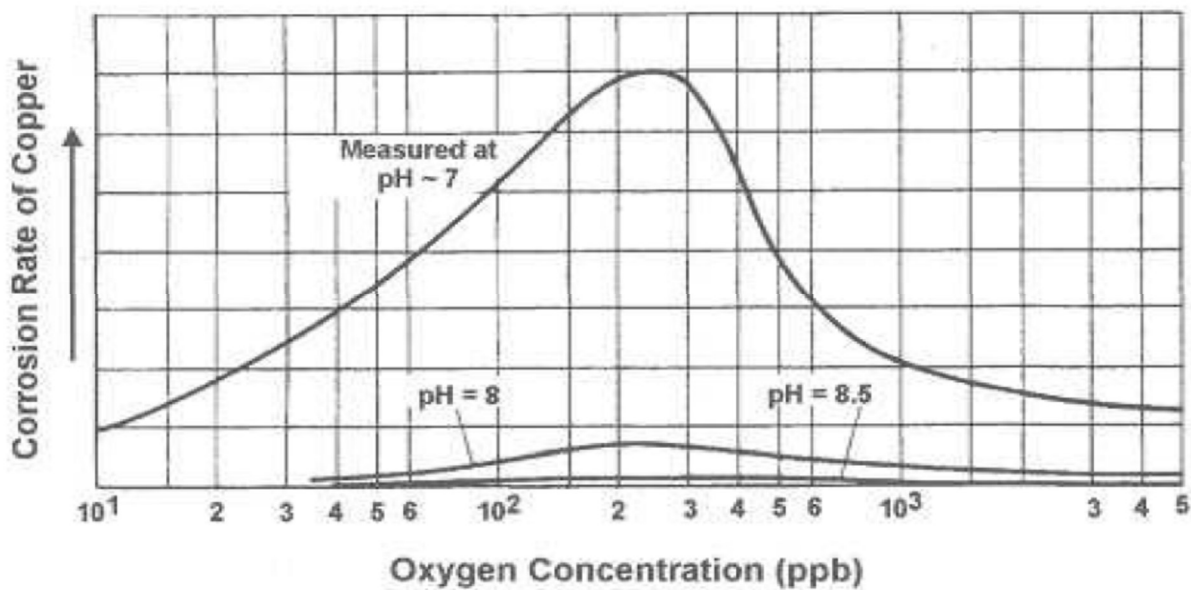
ANNEXURE-7**Schematics of the Mechanism for hollow conductor packing**

- 1 Oxidation of copper
- 2 Release of Oxidised Copper (particle or Ion)
- 3 Transport of the Released Copper
- 4 Re-deposition of the Migrating Copper

ANNEXURE-8

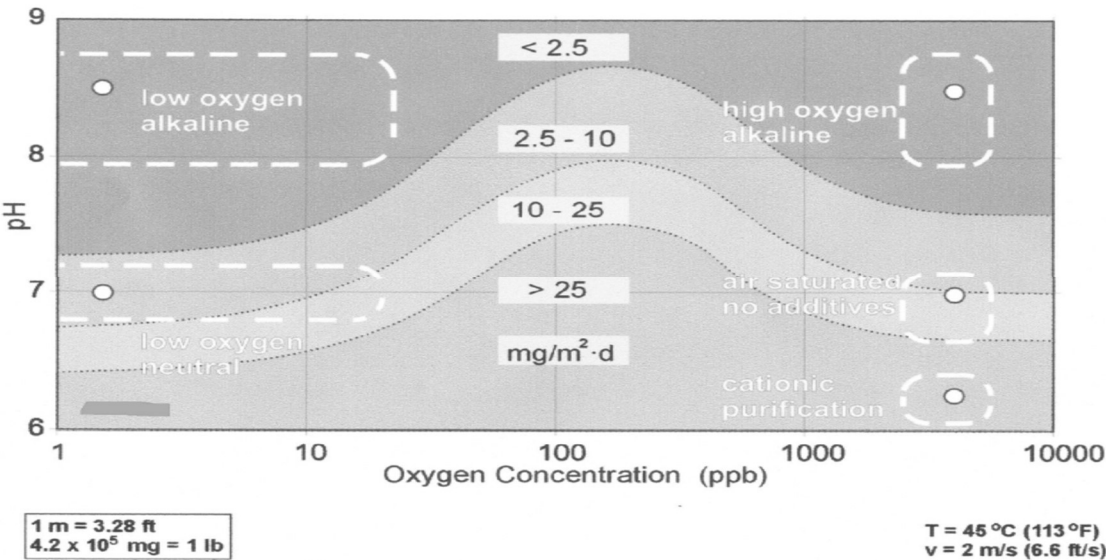
Copper Oxides: Dense Polyhedrons (Left) and Flower-Shaped Needles (Right) formed in High- and Low-Oxygen Water, respectively.

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ANNEXURE-9**COPPER CORROSION RATE V/S OXYGEN CONCENTRATION****ANNEXURE- 10****COPPER CORROSION RATE V/S OXYGEN CONC. AT VARIOUS pH VALUES**

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ANNEXURE-11



**Updated Summary of Copper Release as a Function of Oxygen Concentration and pH;
Operating Areas of the Different Water Chemistry Regimes**